

## Air Quality Health Index (AQHI) — Edmonton

# 2.0

Current AQHI at Gibbons (14.55 km from venue)  
Forecast Sun 12:00 AM: **2.0**

| Health Risk | AQHI | At-Risk Population                   | General Population                        |
|-------------|------|--------------------------------------|-------------------------------------------|
| Low Risk    | 2.0  | Enjoy your usual outdoor activities. | Ideal air quality for outdoor activities. |

No active Environment Canada weather alerts for the venue.

## Environmental Intelligence Summary

- At Edmonton Symphony Orchestra Outdoor Concert, temperature is 19.4°C and feels near 20.0°C. Winds are 10 km/h from S, gusting near 19 km/h.
- The nearest current AQHI is 2 at Gibbons, 14.6 km from the venue.
- The official Government of Alberta AQHI for Edmonton is 4 (forecast tonight: 5, tomorrow: 4).
- A gridded AQHI estimate blending official and community sensors (confidence: High) puts the area near 3.
- The nearest air monitoring station (Southfort, 4.3 km) reports fine particulate matter (PM2.5) at 36.8 µg/m³.
- A nearby community sensor ('AE-Fort Saskatchewan (south)', 5.1 km) reads 28.8 µg/m³ PM2.5.
- Back-trajectory modeling (HRDPS wind fields, 6 h lookback) indicates the air currently at the venue traveled from the vicinity of a point 108.3 km to the S, i.e. arriving from the S.
- The nearest active fire detection (NASA FIRMS – VIIRS, last 2026-08-22) is 11.5 km NE of the venue. The modeled back-trajectory does not pass near this fire (nearest approach 21.8 km), so this specific fire is unlikely to be the source of any smoke reaching the venue right now.
- The AQHI forecast for Sun 12:00 AM is 2.
- Overall operational environmental risk is LOW.

## Recommendations

- Environment Canada AQHI guidance — general population: Ideal air quality for outdoor activities.
- Environment Canada AQHI guidance — at-risk populations: Enjoy your usual outdoor activities.

## Hazard Snapshot — Overall Risk: Low Risk

|                                                                       |                                                            |                                                                            |                                                             |                                                                   |
|-----------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------|
| <b>AIR QUALITY</b><br><b>Low Risk</b><br>2.0 AQHI                     | <b>HEAT</b><br><b>Low Risk</b><br>24.0 °C apparent/humidex | <b>WIND</b><br><b>Low Risk</b><br>19.1 km/h peak gust                      | <b>PRECIPITATION</b><br><b>Low Risk</b><br>0.2 mm/h maximum | <b>THUNDERSTORM</b><br><b>Low Risk</b><br>— first forecast signal |
| <b>AQHI RATE OF CHANGE</b><br><b>Low Risk</b><br>0.0 AQHI change/hour |                                                            | <b>WIND SHEAR</b><br><b>Low Risk</b><br>7.3 ° direction diff (10m vs 120m) |                                                             |                                                                   |

## Local Monitoring

|                                                     |                                                   |                                                             |                                                                                  |
|-----------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------|
| Nearest station<br><b>2.0</b><br>Gibbons · 14.55 km | Blended estimate<br><b>3.0</b><br>confidence High | PM2.5 (station)<br><b>36.8 µg/m³</b><br>Southfort · 4.32 km | PM2.5 (community)<br><b>28.8 µg/m³</b><br>AE-Fort Saskatchewan (south) · 5.12 km |
|-----------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------|

